

The female body



The human body medicine forgot to fully understand

By Jouhaina Kourou



The good physician treats the disease; the great physician treats the patient who has the disease.

- WILLIAM OSLER



But what happens when the patient has rarely been studied?

Dear reader,

Medicine has transformed humanity. Vaccines eradicated diseases. Antibiotics changed history. Artificial intelligence is beginning to reshape healthcare.

However, while reading scientific papers over the past few years, I kept encountering an uncomfortable question.

Why do so many women's diseases remain misunderstood? Why do some women wait years for a diagnosis? Why are certain symptoms dismissed while others are instantly recognized?

This document looks at how science evolved, and how it's evolving now. That matters because the future of healthcare rests on understanding every body, not just the average one.

Welcome.

*With curiosity,
Jouhaina Kourou*

CHAPTER ONE

The woman who was told she had anxiety

She arrived at the emergency department exhausted, nauseated, and with an ache spreading through her jaw and up the back of her neck. Stress, she was told: “Anxiety, maybe. Just try to rest”...

Hours passed. The tests came back. Her heart had been failing the entire time, **“a heart attack”**, unfolding quietly, wearing symptoms **no one had been trained to expect**. She was not imagining it. **She had never been the body medicine was built around.**

One body. Two experiences

OFTEN RECOGNIZED

The classical signs

The textbook heart attack, modeled on male patients

- Crushing chest pain
- Pain radiating down the left arm
- Sudden collapse
- Immediate suspicion of cardiac cause
- **Faster path to diagnosis**

OFTEN MISSED

The overlooked signs

Symptoms women more commonly report

- Extreme fatigue
- Nausea and shortness of breath
- Jaw, neck, and upper-back pain
- Symptoms attributed to stress
- **Longer diagnostic delays**

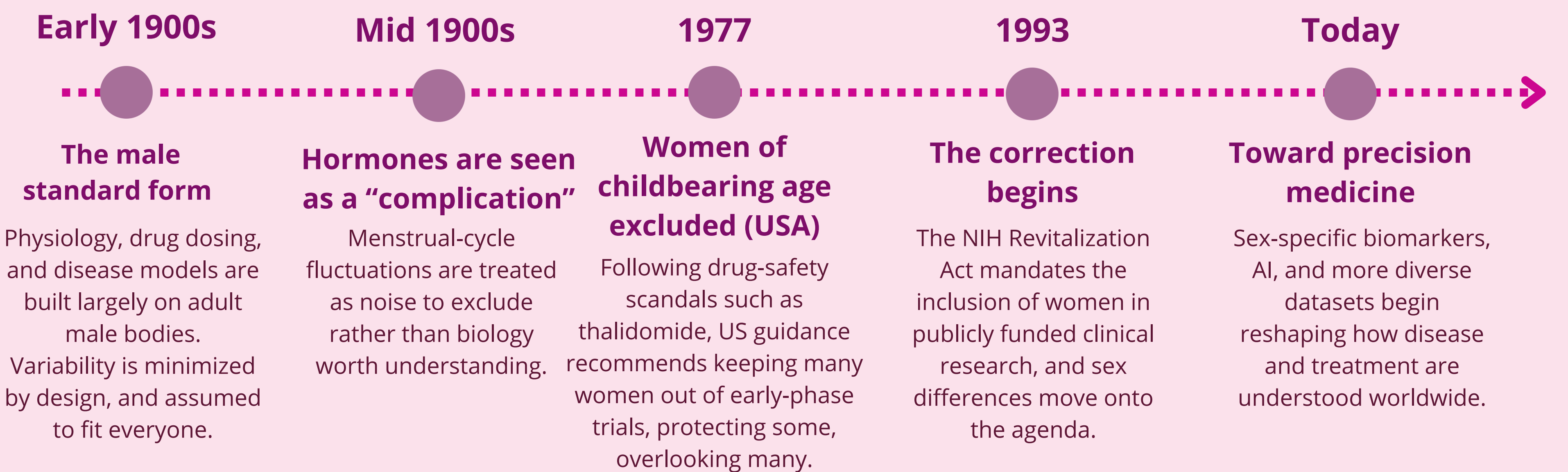
CHAPTER TWO

How men accidentally became medicine's default?

For much of the twentieth century, the "standard" research subject was an adult man. Trials enrolled mostly male participants; labs relied on male animals; whole physiological models were drawn from male data, and assumed to fit everyone.



A century of the “Average male”



“

Women aren't excluded because they matter less. They were excluded because science underestimated how much sex differences matter.

CHAPTER THREE

The diseases that changed the conversation

Four conditions exposed the gaps in what medicine thought it knew. Each one begins with a different question.

THE LEADING CAUSE

Heart disease

Imagine running a marathon while breathing through a straw.
That's how some women describe a heart attack.

Heart disease remains the leading cause of death among women worldwide. For decades, it was framed as a man's disease. The "classic" symptoms everyone learnt to watch for were those that men tend to have.

Women's hearts don't function differently. Our early understanding of them was simply built on predominantly male populations, so the signals women more often experience were treated as atypical, and too often, missed.

#1

Cause of death in women
worldwide

Atypical

Female symptoms are often
labeled "unusual"

Delayed

Diagnosis can come later than
for men

DECADES OF SILENCE

Endometriosis

What if pain became so normal that everyone stopped asking why?

Despite affecting millions of women worldwide, endometriosis has historically taken years to diagnose. Many patients are told that severe menstrual pain is simply "normal."

In fact, it isn't. Painful periods are common, but pain severe enough to interrupt work, school, or daily life deserves investigation. Only recently has the condition begun receiving the research attention it has long deserved.

Millions

Affected worldwide

Years

Typical delay to diagnosis

“Normal”

The word that hid it for
decades

A CASCADE EFFECT

PCOS

One hormone. Many consequences.

Polycystic ovary syndrome is one of the most common hormonal conditions in women, and one of the most under-recognized. Because its effects ripple across metabolism, fertility, skin, mood, and long-term health, it is easy to treat each symptom in isolation and miss the pattern connecting them.

Understanding how a single hormonal imbalance can shape so many systems is exactly the kind of question that sex-focused research is now built to answer

Common

Among hormonal conditions
in women

Systemic

Metabolic, reproductive, and
more

Under-diagnosed

Frequently recognized late

AN OPEN QUESTION

Autoimmune diseases

Why does the immune system attack women for more often than men?

Roughly four out of five people living with autoimmune diseases are women. **Lupus, rheumatoid arthritis, multiple sclerosis, and Sjögren's syndrome** all disproportionately fall on **female bodies**.

Genetics, hormones, environmental exposures, and immune regulation all likely play a role, but the full answer remains one of medicine's most compelling unsolved questions, and a reason sex differences can no longer be treated as a footnote.

4 in 5

Autoimmune patients are women

Many

Conditions, one shared skew

Unsolved

The “why” is still open

FACT OR MYTH?

“Women are simply smaller men.”



Women are not scaled-down versions of men. Differences in body composition, hormones, liver enzymes, kidney function, and immune response change how medications are absorbed, distributed, metabolized, and eliminated.

That's why some drugs have later been found to need different dosing or carry different side-effect profiles in women, and why testing treatments across diverse bodies isn't optional. It's how medicine gets the answer right.

DID YOU KNOW?

Women were historically underrepresented in many clinical drug trials.

For decades, pregnant women in particular were among the most frequently excluded groups, leaving clinicians to make difficult decisions with incomplete evidence about which treatments are safest during pregnancy. Keeping patients safe and actually studying them were never meant to be in conflict.

N U M B E R S T H A T M A T T E R

The scale of the gap

Millions of women worldwide are affected by endometriosis, a condition still routinely diagnosed years too late.

~4/5 of people living with autoimmune diseases are women, for reasons science is still working to explain.

Years of diagnostic delay reported in many studies for conditions primarily affecting women.

CHAPTER FOUR

The invisible bias

The next gap isn't in biology but in perception. How pain is heard, how symptoms are described, and what a "typical" patient is assumed to look like all shape care long before any test is run.

When pain isn't believed

Pain is one of medicine's most subjective symptoms, and one of its most revealing.

Studies suggest that women presenting with pain may, on average, wait longer for pain relief in some emergency settings and are more likely to have their symptoms attributed to stress or psychological causes. This doesn't happen everywhere, or to everyone, but the pattern is documented often enough to matter.

What makes this bias so stubborn is that it's rarely deliberate. It lives in language, in stereotypes, in medical education, and in decades-old assumptions about who feels what. It can shape the judgment of any clinician, including women physicians, precisely because no one chose it.

Recognizing that is not an accusation. It's the first step in correcting it and one of the most fascinating places where psychology and medicine meet.

CHAPTER FIVE

Can algorithms succeed where medicine fell short?

Artificial intelligence can see patterns across millions of bodies at once. The question is whether it will correct medicine's blind spots, or quietly inherit them.

The promise, and the catch

The future of women's health may be shaped by algorithms. AI is already being explored to discover sex-specific biomarkers, improve early diagnosis of conditions like endometriosis, predict pregnancy complications, personalize drug dosing, and map how diseases progress differently in women and men.

Add digital twins, wearables, and precision-medicine models, and you get something medicine has never had. The ability to ask not, "What works for most patients?" but "What works for this one?"

But an algorithm is only as good as its data.

If historical datasets underrepresent women, AI risks automating the very bias it could help erase, encoding yesterday's blind spots into tomorrow's tools. Building datasets that reflect diverse bodies isn't a technical detail. It's the whole game.

CHAPTER SIX

What the future looks like?

Imagine walking into a hospital twenty years from now. An AI system instantly reads your genetics, hormones, microbiome, lifestyle, and history, and instead of asking "*what works for most patients?*", it asks "*what works for you?*"

That future begins with one simple idea “**there is no default human body. Only individual ones.**”

Understanding women better
doesn't only improve women's
healthcare, it also improves
medicine itself.
